INCH-POUND
MIL-DTL-28840/15E
w/AMENDMENT 3
09 September 2022
SUPERSEDING
MIL-DTL-28840/15E
w/AMENDMENT 2
03 March 2021

DETAIL SPECIFICATION SHEET

CONNECTORS, ACCESSORIES, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, PROTECTIVE COVER, PLUG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.

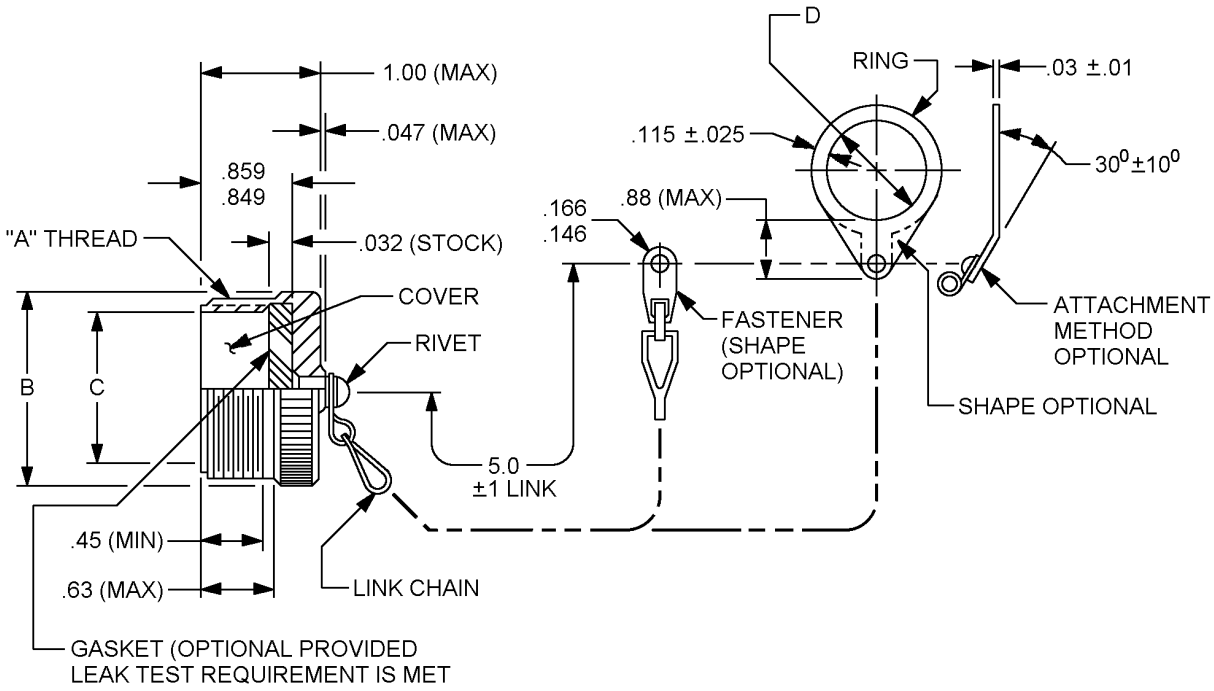


FIGURE 1. Dimensions and configurations.

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Shell size designator (see note 1)	A thread class 2B	B max dia	C ± .006 (± 0.15)	D ± .010 (± 0.25)
A (11)	.750-1P-.2L-DS	1.028 (26.21)	.575 (14.60)	.765 (19.43)
B (13)	.875-1P-.2L-DS	1.141 (28.98)	.699 (17.75)	.890 (22.61)
C (15)	1.062-1P-.2L-DS	1.263 (32.08)	.871 (22.12)	1.015 (25.78)
D (17)	1.125-1P-.2L-DS	1.387 (35.23)	.941 (23.90)	1.140 (28.96)
E (19)	1.312-1P-.2L-DS	1.513 (38.43)	1.108 (28.14)	1.265 (32.13)
F (23)	1.500-1P-.2L-DS	1.703 (43.26)	1.293 (32.84)	1.453 (36.91)
G (25)	1.625-1P-.2L-DS	1.825 (46.35)	1.454 (36.93)	1.577 (40.06)
H (29)	1.812-1P-.2L-DS	2.143 (54.43)	1.620 (41.15)	1.890 (48.01)
J (33)	2.000-1P-.2L-DS	2.329 (59.16)	1.812 (46.02)	2.077 (52.76)

Inches	mm	Inches	mm
.01	0.25	.450	11.43
.025	0.64	.630	16.00
.03	0.76	.849	21.56
.032	0.81	.859	21.82
.047	1.19	.88	22.35
.115	2.92	1.00	25.40
.146	3.71	5.00	127.00
.166	4.22		

NOTES:

1. Shell sizes are provided within parentheses for information and are not a part of the designator.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Dimensions apply after plating.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.
Interface dimensions shall conform to MIL-DTL-28840.
Chain shall be free to rotate on rivet.
Chain material: 300 series corrosion resistant steel, passivated in accordance with SAE-AMS2700.
Gasket shall be mechanically or chemically captive to the cover.
Gasket material: Fluorosilicone.
Material and finish, protective covers: See table I.

TABLE I. Material and finish.

Material and finish designator	Material	Finish
W	Aluminum alloy in accordance with ASTM B85/B85M, ASTM B26/B26M, ASTM B108/B108M, ASTM B209, ASTM B211 or ASTM B221	Cadmium plate, olive drab over electroless nickel
D	Corrosion resistant steel in accordance with ASTM A276, ASTM A473, or ASTM A580/A580M, type 316	Cadmium plate, black
F	Aluminum alloy in accordance with ASTM B85/B85M, ASTM B26/B26M, ASTM B108/B108M, ASTM B209, ASTM B211 or ASTM B221	Pure electro-deposited aluminum, non-reflective
L		Tin-zinc, non-reflective
S		Zinc nickel, non-reflective

Marking: The cover shall be identified with the manufacturer's name or trade-mark and the Part or Identifying Number (PIN).

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PIN example:

M28840/ 15 H A W

M28840/	Military designator
15	Specification sheet number
H	Shell size designator
A	Chain attachment designator
	A – Link chain with fastener
	B – Link chain with ring
W	Material and finish designator
	W – Aluminum, cadmium plate over electroless nickel
	D - Corrosion resistant steel, cadmium plate, black
	F - Wrought aluminum, pure electrodeposited aluminum - non-reflective
	L - Wrought aluminum, tin-zinc - non-reflective
	S - Wrought aluminum, zinc nickel - non-reflective

RESTRICTIONS:

Material and finish designators W and D are inactive for NAVSEA new design, unless approved by NAVSEA.

QUALIFICATION: One protective cover of each shell size for which qualification is desired shall have the following tests performed in the sequence shown:

Visual and mechanical.

Magnetic permeability.

Shell conductivity.

Corrosion.

Air leakage: Cover shall be mated to suitable connector plug and tested in accordance with test procedure EIA-364-02. The following details shall apply: The air leakage rate shall not be greater than one atmospheric cubic inch per hour (4.55×10^{-3} cubic centimeter per second) at a differential pressure of at least 10 psi.

Tensile strength: Cover shall be mounted to a suitable fixture to facilitate a 25-pound pull applied at the rate of 5 pounds per second and held for a period of 5 minutes. The load shall be applied to the chain end fastener through the center line axis of the protective cover, and then repeated through a perpendicular axis (90°) from the centerline axis. Chain or fastener breakage, or loosening of the rivet, shall be considered failures.

Shell conductivity.

Post visual and mechanical.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to MIL-DTL-28840, this document references the following:

ASTM A276
ASTM A473
ASTM A580/A580M
ASTM B26/B26M
ASTM B85/B85M
ASTM B108/B108M
ASTM B209
ASTM B211
ASTM B221
EIA-364-02
SAE-AMS2700

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2022-176)

Review activities:
Army – AV, MI
Navy - OS, SH

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